Universal Motives for Relationships & Sex

An Exploration of Evolutionary & Biosocial Perspectives

This lecture will examine the fundamental evolutionary mechanisms that shape human mating motivations and behaviors. We will explore how motives for sexual desire and romantic love evolved, how individuals select mates based on perceived value, and how social and cultural environments interact with these biological predispositions. We will deconstruct the processes of mate selection, long-term strategies, and the mechanisms, such as jealousy, that have evolved to guard established relationships.

Our analysis will progress through five core conceptual areas:

1. **Sexual Motivation & Selection**
2. **Mate Value & Genetics**
3. **Parental Investment & Long-Term Preferences**
4. **Biosocial Theory & Culture**
5. **Relationship Guarding & Jealousy**

1. Sexual Motivation & Selection
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3. Parental Investment & Long-Term Preferences
4. Biosocial Theory & Culture
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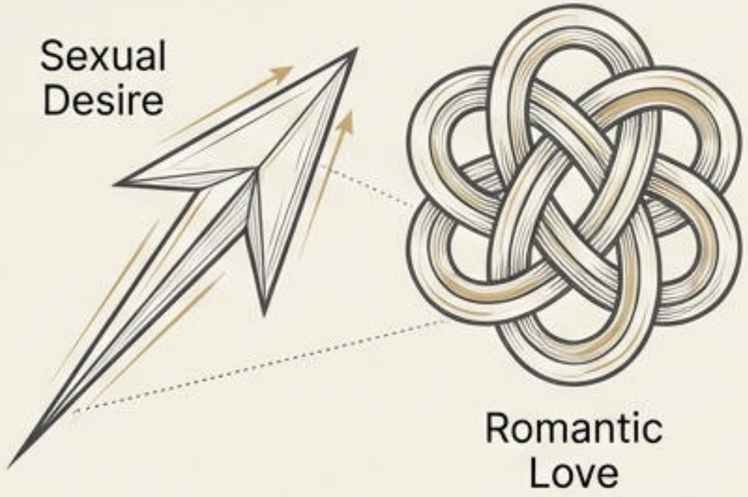
Sexual Motivation & Selection



Distinct Motives: Sexual Desire vs. Romantic Love

Two separate and independent universal motives have evolved to ensure the continuation of the human species: **sexual desire** and **romantic love**. It is critical to understand their distinct functions.

- **Sexual Desire**: This motive evolved to facilitate the sexual behavior necessary for conception. Without it, the time, effort, and potential risks of sexual intercourse might be perceived as prohibitive. Its primary evolutionary function is to initiate the process of reproduction.

- **Romantic Love**: This emotion refers to the strong attachment and bond individuals form, promoting long-term commitment. Its evolutionary benefits are multifaceted:
  - **Mate Retention**: It serves as the ‘glue’ that binds a couple, reducing the motivation to search for alternative partners and helping to prevent desertion.
  - **Division of Labor**: It promotes the cooperative division of labor essential for this survival and health of the couple and, crucially, their offspring.
  - **Emotional Well-being**: The presence of a partner elicits pleasant feelings, while their absence causes anguish and distress, reinforcing the bond.

Evidence for the universality of romantic love was documented by Jankowiak and Fischer (1992), who found its presence in 88.5% of 166 societies studied.

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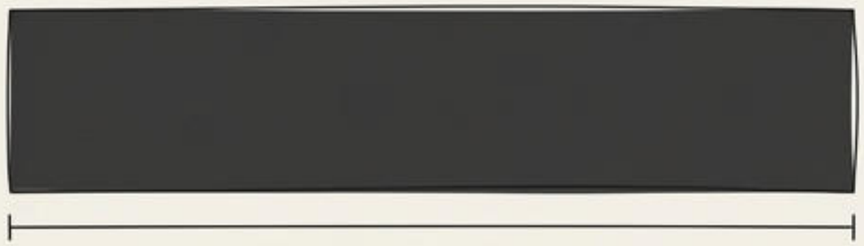
The Role of Orgasm & The ‘Orgasm Gap’

In addition to sexual desire, orgasm functions as a powerful incentive and reinforcer for sexual intercourse. However, its consistency and evolutionary interpretation differ significantly between the sexes.

Male Experience

For males, orgasm is almost invariably associated with ejaculation, providing a consistent and powerful reinforcement for the sexual act that leads to potential conception. Men achieve orgasm nearly 100% of the time during intercourse.



Male Orgasm Consistency, 100%

Female Experience & The ‘Orgasm Gap’

The reward for women is notably inconsistent. A meta-analysis by Lloyd (2005) of 32 reports shows a wide range of frequency:

- **25%** always have an orgasm during intercourse.
- **55%** have one more than half the time.
- **23%** sometimes do.
- **33%** rarely or never do.
- **5-10%** have never had one.

Female Orgasm Frequency (Lloyd, 2005)

Despite this inconsistency, women still derive significant benefits from intercourse, including feelings of excitement, intimacy, and emotional closeness with their partners, which helps to reinforce the pair-bond central to romantic love.

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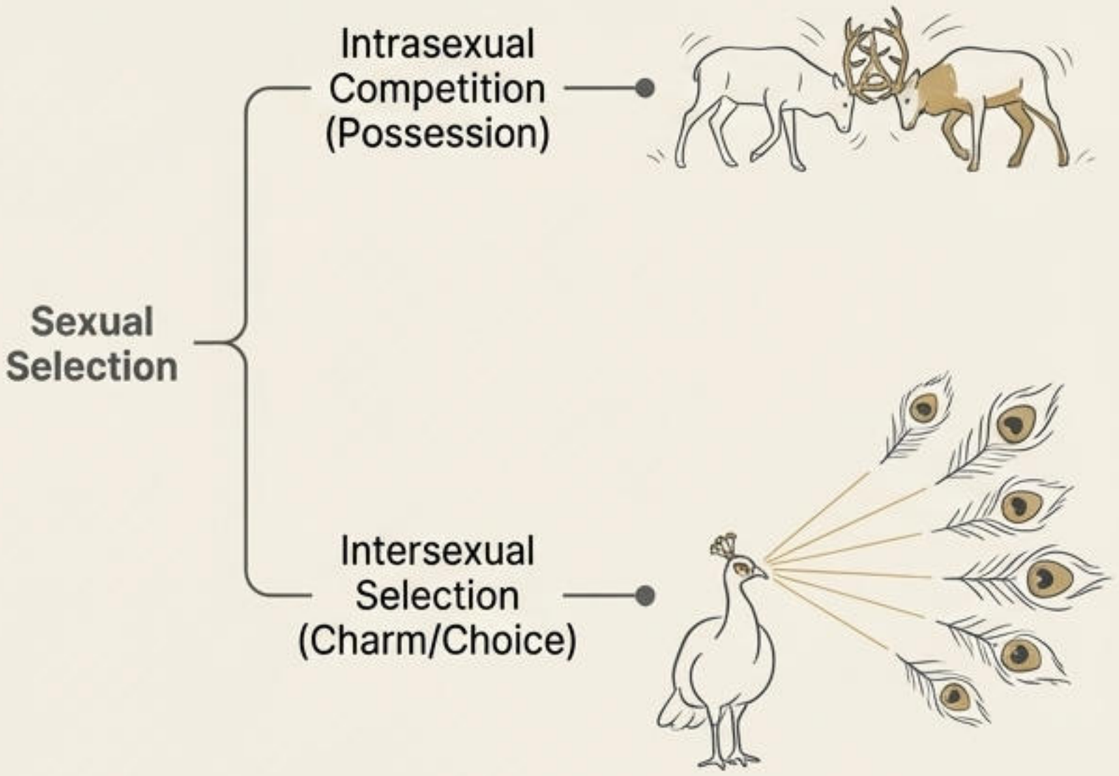
Darwin’s Concept of Sexual Selection

The fulfillment of sexual desire and romantic love requires a cooperative partner. The process of choosing that partner is governed by **sexual selection**, a concept introduced by Charles Darwin.

Natural selection concerns survival, but for a species to benefit, adaptive traits must be passed to subsequent generations. This necessitates cooperation for conception (short-term) and infant care (long-term).

Darwin defined sexual selection as the "struggle between males for possession of the females." This struggle can manifest in two primary ways:

1. **Intrasexual Competition (Possession):** Males compete directly with other males, often through aggression or combat, to gain exclusive access to females. The winner secures mating opportunities.
2. **Intersexual Selection (Charm/Choice):** Males compete to be chosen by females. In this case, the female acts as the selecting agent. Her preferences for certain male traits (e.g., charm, appearance) determine who is allowed to mate. A classic example is the peahen, who selects peacocks based on the number of eyespots on their tail feathers, which correlates positively with copulation success.



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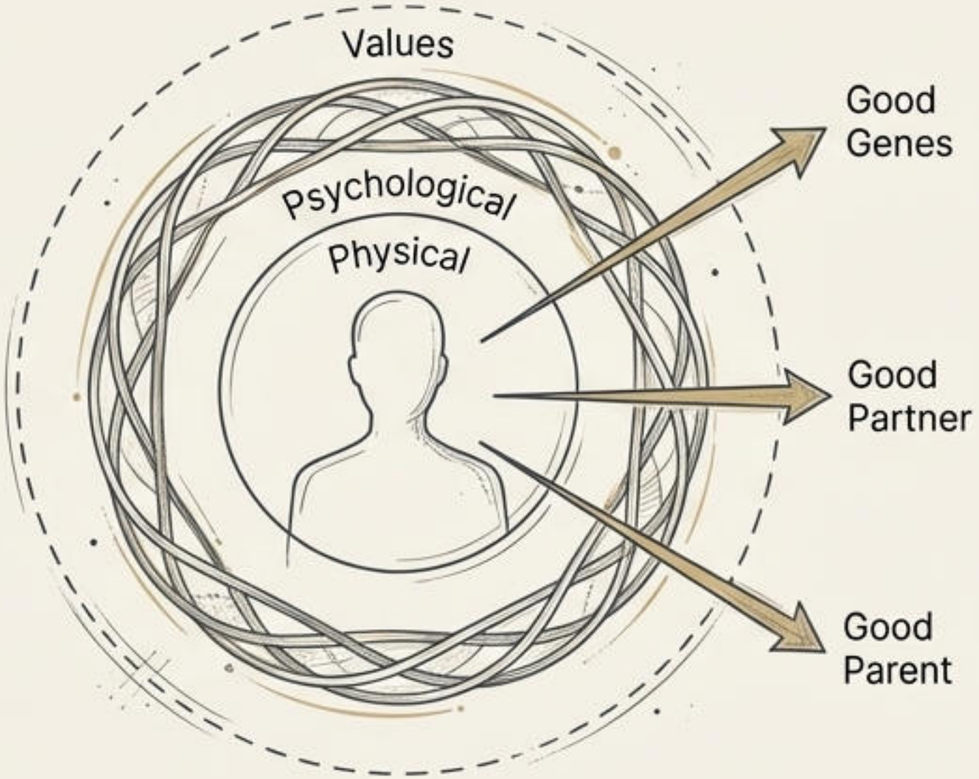
Mate Value & Genetics

Defining Mate Value

A person's **mate value** refers to their overall **attractiveness to the opposite sex**, based on the possession of a constellation of desired characteristics. A higher mate value corresponds to greater appeal as a potential partner. These characteristics can be categorized into:

- **Physical Features:** Traits like an attractive face and body are immediately apparent but can fade over time. They are subject to habituation.
- **Psychological Characteristics:** Traits such as ambition, a sense of humor, loyalty, and emotional stability take time to discover but are often more enduring.
- **Value System**: Shared interests and core values significantly increase mutual attraction and compatibility.

Collectively, these traits signal an individual's quality in three key domains: having good genes, being a good partner, and being a good parent.



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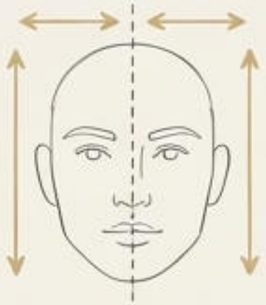
The Good Genes Hypothesis & Markers of Attractiveness

The preference for attractive partners is a human universal, believed to have arisen from sexual selection. The **good genes hypothesis** posits that physical attractiveness serves as an honest signal for underlying genetic quality, specifically health, fertility, and intelligence.

Key universally recognized markers of facial attractiveness include:

- **Symmetry:** The degree to which the right and left halves of the face match. Asymmetry can be a subtle indicator of developmental instability or poor health.
- **Sexual Dimorphism:** The extent to which a face exhibits typical masculine or feminine features. More masculine male faces and more feminine female faces are generally rated as more attractive.
- **Averageness:** Faces that represent a composite or 'average' of many facial configurations within a population are often perceived as more attractive, as this may signal a lack of genetic abnormalities.

While 'average' is a baseline, highly attractive faces often possess exaggerated ideal features, such as redder lips and darker eye lines in females, or a darker brow and more beard growth in males, compared to the population average.



Symmetry

Masculine Feminine

Sexual Dimorphism

Averageness

Composite signals genetic stability.

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Empirical Link: Mate Value & Mating Success

If attractiveness is a key component of mate value, it should predict an individual's success in attracting partners. Research by Rhodes et al. (2005) in an Australian sample provides direct evidence for this link.

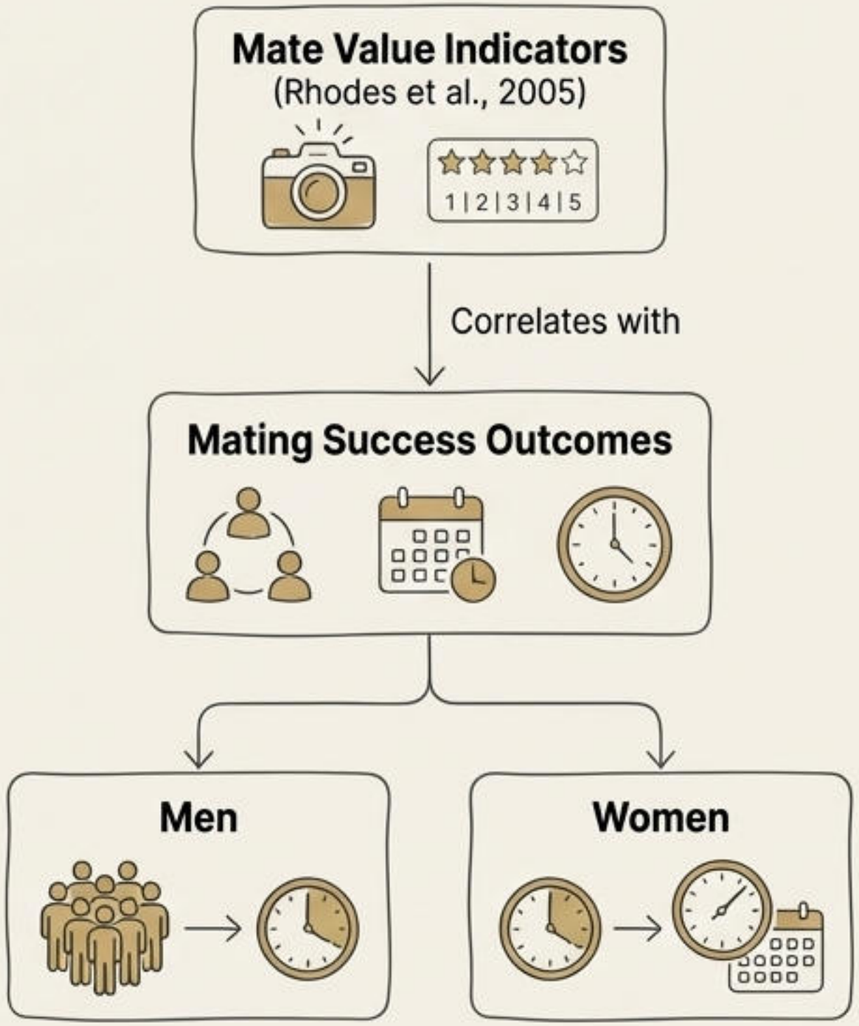
The study measured:

- **Mate Value Indicators:** Photos of participants were rated on attractiveness, sexual dimorphism, averageness, and symmetry.
- **Mating Success Outcomes:** Participants self-reported their number of sexual partners, age of first intercourse, and relationship durations.

Findings:
The results confirmed a significant correlation. Individuals with higher physical mate values reported more extensive relationship experiences.

- **For Men:** Those with more attractive faces and attractive, masculine bodies had more sexual partners and initiated sexual activity at an earlier age.
- **For Women:** Those with more attractive faces also became sexually active at an earlier age and had more long-term relationships (exceeding 12 months).

This provides empirical support for the idea that physical traits associated with high mate value translate into tangible mating advantages.



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Empirical Link: Attractiveness & Reproductive Success

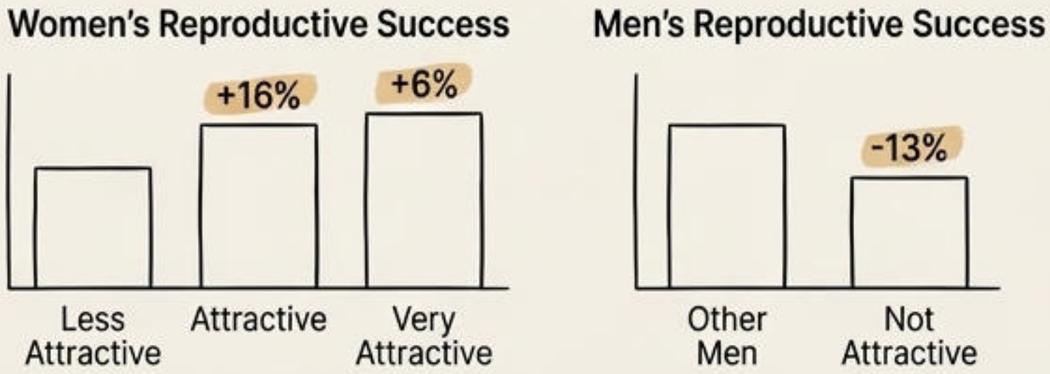
Beyond simply attracting partners, does attractiveness predict ultimate reproductive success—the number of children one has? Several studies suggest it does, supporting the evolutionary interpretation that beauty signals reproductive capacity.

Pflüger et al. (2012):

- Male students rated photos of 20-year-old women for attractiveness.
- Later in life, the number of children these women had was counted.
- **Result:** Women rated as more attractive and sexier when young had more pregnancies and gave birth to more children (this correlation held for women not using contraception).

Jokela (2009):

- Yearbook photos of over 2,000 individuals were rated for attractiveness.
- This was correlated with the number of children they had 35-38 years later.
- **Result:** “Attractive” women had 16% more children and “very attractive” women had 6% more children than their less attractive peers. “Not attractive” men had 13% fewer children than other men.



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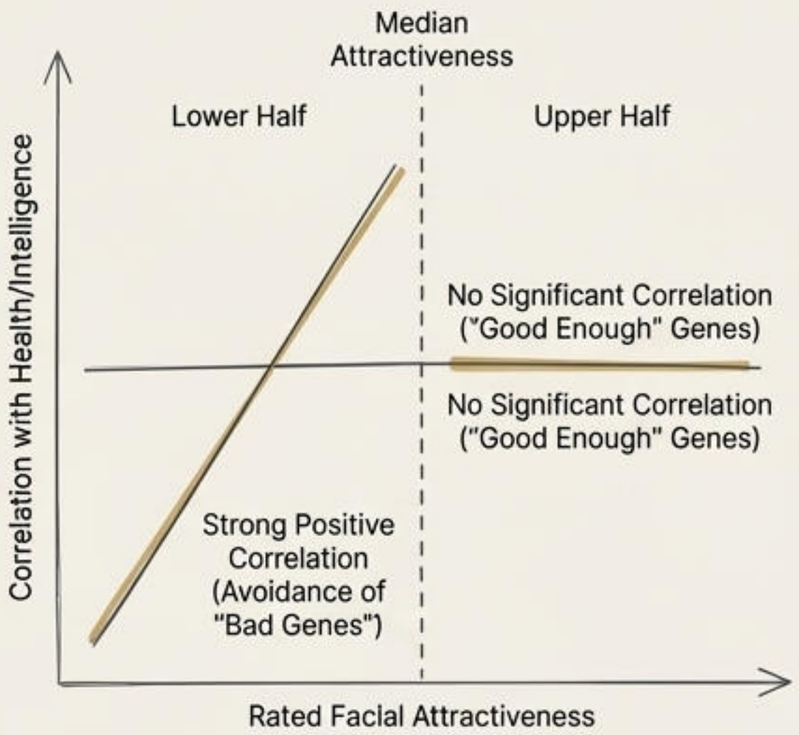
The Inverse: The Bad Genes Hypothesis

An alternative but complementary perspective to the good genes hypothesis is the **bad genes hypothesis. This theory proposes that mate selection is driven less by attraction to *exceptionally* good genes and more by a strong motivation to *avoid* individuals whose appearance signals bad genes** (e.g., potential for disease, low intelligence).

Zebrowitz & Rhodes (2004) Study: This longitudinal study analyzed facial photos, intelligence scores, and health data from individuals at multiple life stages. They divided faces at the median level of attractiveness. The stages. They divided faces at the median level of attractiveness. The hypothesis was that correlations between unattractiveness, poor health, and low intelligence would be strongest ***below*** the median. Above the median, all individuals are presumed to have “good enough” genes.



Findings: The results supported the hypothesis. A positive correlation between facial attractiveness, health, and intelligence was found ***only for individuals in the lower half*** of attractiveness ratings. For these individuals, as rated attractiveness decreased, so did their measured health and intelligence. This suggests a crucial function of mate selection is to screen out partners who fall below a minimum threshold of health and fitness signals.

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Parental Investment & Long-Term Preferences

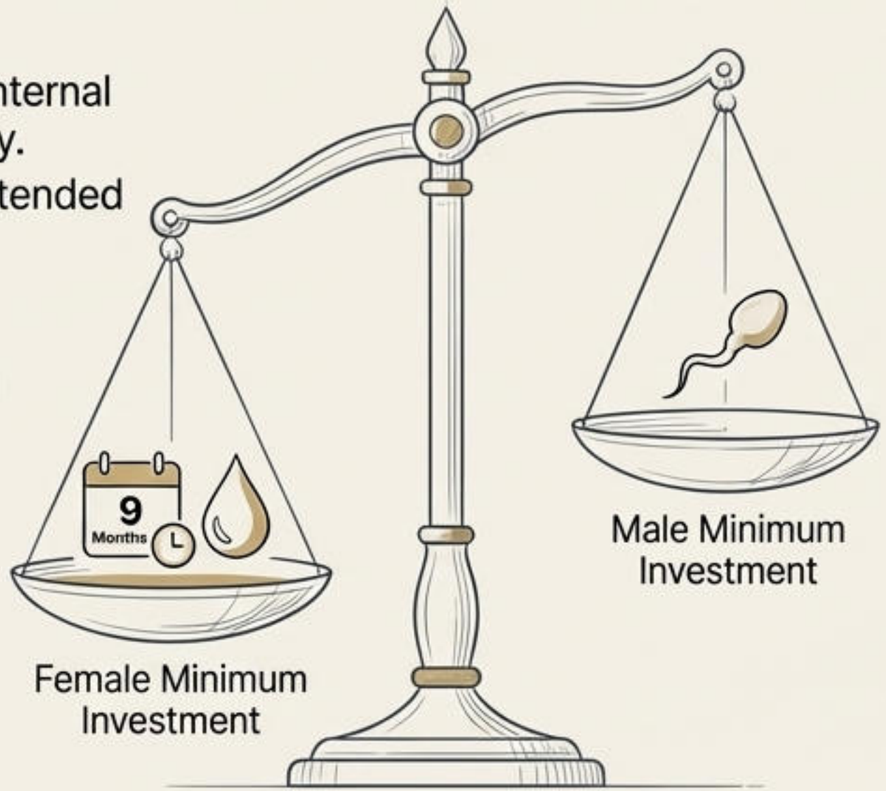
Parental Investment Theory: The Fundamental Asymmetry

For long-term mating, evolutionary success depends not just on conception but on raising offspring to maturity. The biological differences between men and women create a fundamental asymmetry in the minimum obligatory parental investment required to produce a child.

Female Investment (High):

- **Gestation:** A 38-week (9-month) internal investment of resources and energy.
- **Lactation:** Historically, a further extended period of providing nourishment.
- **Reproductive Rate:** A woman's reproductive capacity is finite, limited by age and the long interval between births. Each child represents a massive investment.

This biological asymmetry is the theoretical foundation for predicting different long-term mating strategies and preferences between the sexes.



Male Investment (Low):

- **Minimum Obligation:** The act of insemination.
- **Reproductive Rate:** A man's potential reproductive rate is vastly higher, limited primarily by the number of fertile partners he can access.

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Sex-Differentiated Preferences in Long-Term Mates

The asymmetry in parental investment leads to distinct evolutionary "problems" for each sex to solve, resulting in different preferences for characteristics in a long-term partner.

Women's Strategy & Preferences:

- Problem: Securing resources and commitment to ensure the survival and success of her highly valuable offspring.
- Predicted Preference: Women are predicted to place a higher value on characteristics in a man that signal the ability and willingness to provide long-term investment. This includes:
  - Resources: Good financial prospects.
  - Status & Ambition: industriousness and social standing.
  - Commitment: Cues of emotional stability and dependability.

Problem: Secure Investment

Solution: Value Resources/Commitment

Men's Strategy & Preferences:

- Problem: Identifying fertile partners to maximize reproductive success.
- Predicted Preference: Men are predicted to place a higher value on characteristics in a woman that signal high reproductive and fertility potential. This includes:
  - Youth: Age is a strong correlate of fertility.
  - Health & Fecundity: Physical attractiveness ('good looks') serves as an honest signal of health and ability to bear healthy children.

Problem: Identify Fertility

Solution: Value Youth/Health

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Cross-Cultural Evidence: The Buss (1989) Landmark Study

To test if these predicted sex differences in mate preferences are human universals, David Buss (1989) conducted a large-scale survey across 37 cultures in 33 countries. The results provided powerful support for the parental investment theory.

Key Universal Findings:

- **Women Valued Resources More:** In nearly all cultures, women rated **good financial prospects** and **ambition/industriousness** as more important in a long-term mate than men did. These traits signal a man's capacity to be a reliable provider.
- **Men Valued Youth & Attractiveness More:** Overwhelmingly, men in all cultures rated **good looks** (physical attractiveness) as more important in a mate than women did. Men also consistently preferred younger partners. These traits are reliable cues to a woman's fertility and reproductive value.

The consistency of these findings across diverse cultures strongly suggests an evolved, species-typical psychological architecture for mate selection rather than a purely cultural artifact.

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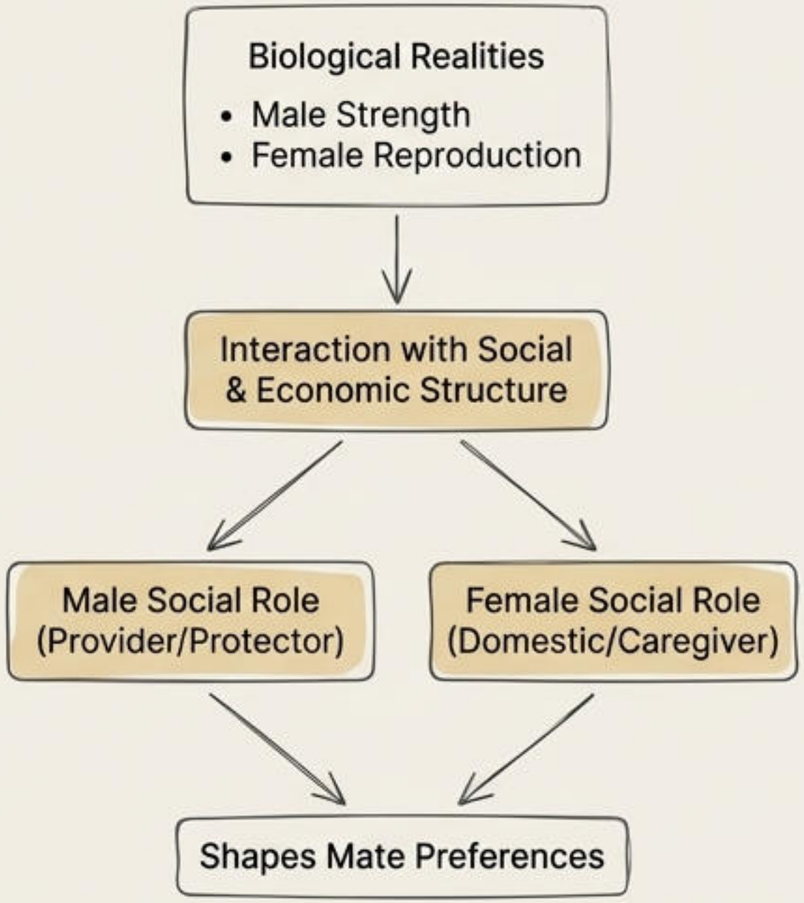
Biosocial Theory & Culture

An Alternative Perspective: Biosocial Theory

While evolutionary psychology emphasizes our ancestral past, biosocial theory (Wood & Eagly, 2002) provides a framework that stresses the interaction between **evolved biological differences and the current social environment. The theory's foundation rests on two undisputed biological sex differences:**

1. Strength: Men are, on average, physically larger and stronger.
2. Reproduction: Only women can bear and nurse children.

According to biosocial theory, these biological realities interact with a society's economy, social structure, and technology to shape society's economy, social structure, and technology to shape gender roles. Societal expectations arise where men, due to their strength, assume roles of provider and protector, while women, due to their reproductive capacity, assume domestic and childcare roles. These socially constructed roles, not evolved psychological mechanisms, are seen as the primary drivers of psychological gender differences in mate preferences.



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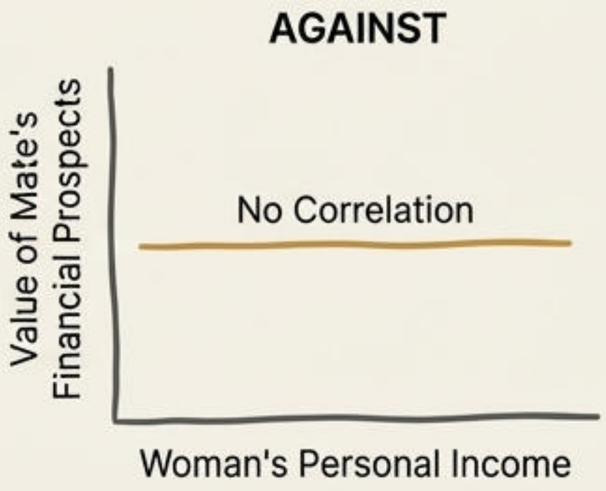
The Structural Powerlessness Hypothesis

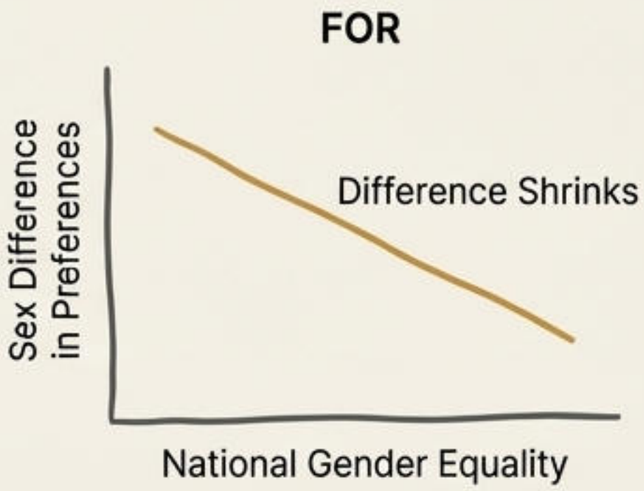
Biosocial theory offers a direct challenge to the evolutionary explanation for women's preference for resourceful men. The structural powerlessness hypothesis argues that:

- Men and women desire the same financial resources.
- However, societal structures and historical discrimination have made it more difficult for women to achieve economic power and status independently.
- Therefore, women seek these resources through marriage, by selecting a male partner who already possesses them. It is a pragmatic strategy to overcome economic inequality, not an innate preference.

Conflicting Evidence:

- **Against the Hypothesis** (Wiederman & Allgeier, 1992): Found no relationship between a woman's personal income and the importance she placed on a mate's financial prospects. High-earning women valued it just as much as low-earning women.
- **For the Hypothesis** (Eagly & Wood, 1999): Re-analyzing Buss's (1989) data, they found that in countries with greater gender equality, the sex differences in preferences shrink. As women's economic power increases, their preference for a spouse's income potential decreases, and men's preference for domestic skills also decreases.





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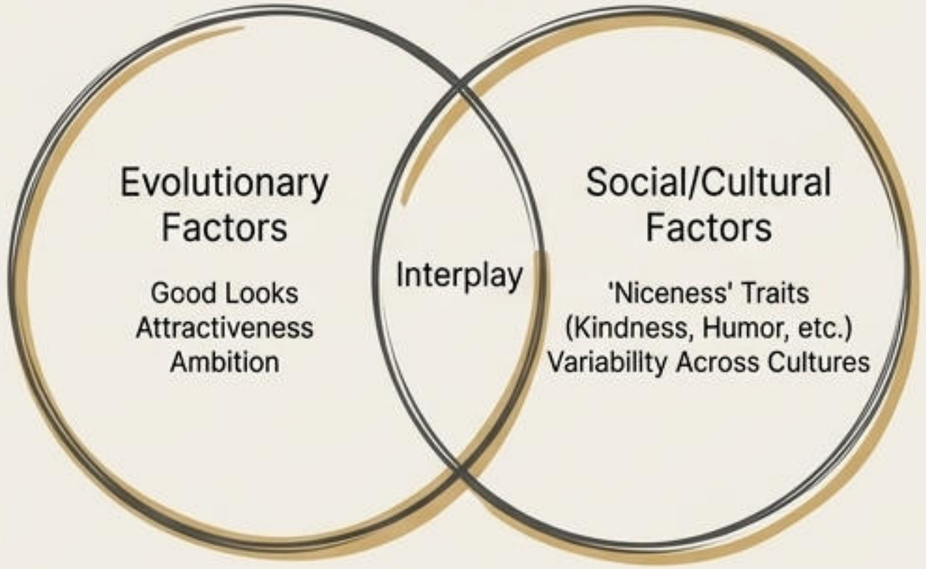
The BBC Worldwide Survey: Evolved & Social Factors

Further evidence for the interplay of evolution and culture comes from a large-scale BBC survey (Lippa, 2007) with over 200,000 participants from 53 countries. The top nine most important traits for a long-term mate were (in order): intelligence, humor, honesty, kindness, good looks, facial attractiveness, values, communication skills, and dependability. The results showed a dual pattern:

1. Support for Evolutionary Predictions:
   - **Good Looks/Attractiveness:** In all 53 countries, men rated good looks and facial attractiveness as more important than women did, replicating Buss's findings.
   - **Ambition:** Women rated ambition as more important than men did, also consistent with evolutionary models.
2. Support for Social/Cultural Factors:
   - **'Niceness' Traits:** Women showed a stronger preference for a cluster of traits Lippa termed 'niceness' (communication skills, dependability, kindness, humor, honesty).
   - **Variability:** Unlike the universal preference for looks, the magnitude of the sex difference in 'niceness' varied widely across countries, suggesting a strong cultural influence.



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The Role of Gender Equality & ‘Luxury’ Traits

Lippa’s (2007) analysis of the BBC data revealed a critical pattern related to socio-economic conditions. The preference for “niceness” was not random; it correlated with national development and gender equality.

Key Finding:
In more economically developed countries with greater gender equality, *both* men and women placed a higher value on “niceness” traits in a partner.

Interpretation:
This suggests that niceness traits (kindness, humor, communication) function as a kind of “luxury good” in the mating market.

- In **less developed nations** with greater inequality, traits indicating survival and provision (ambition, money) are **necessities**. Mating choices prioritize these fundamental needs.
- In more **developed**, egalitarian nations, where basic material needs are more easily met by both sexes, individuals can ‘afford’ to prioritize psychological and emotional qualities that contribute to relationship satisfaction.

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Relationship Guarding & Jealousy

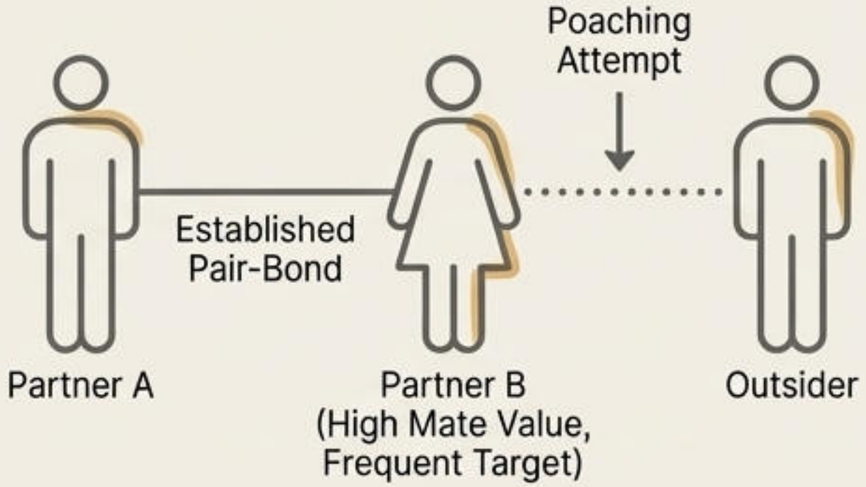
Threats to the Pair-Bond: Mate Poaching

Once a relationship is formed, it is not guaranteed to last. A significant threat is **mate poaching**, defined as the attempt to attract an individual who is already in a committed romantic relationship for either a short-term or long-term liaison.

Global Prevalence (Schmitt et al., 2004): Mate poaching is a common mating strategy observed worldwide. A survey of nearly 17,000 individuals across the globe found that:

- Nearly 60% of men reported having made an attempt to poach someone's partner.
- Nearly 40% of women reported the same.

Targeting High Value Individuals: According to Buss (2007), individuals with high mate value are disproportionately the targets of poaching. Because desirable partners are a scarce resource and often already in relationships, poaching becomes a viableg becomes a viable, if risky, strategy for acquiring a high-quality mate. This persistent threat created the selective pressure for an evolved defense mechanism.



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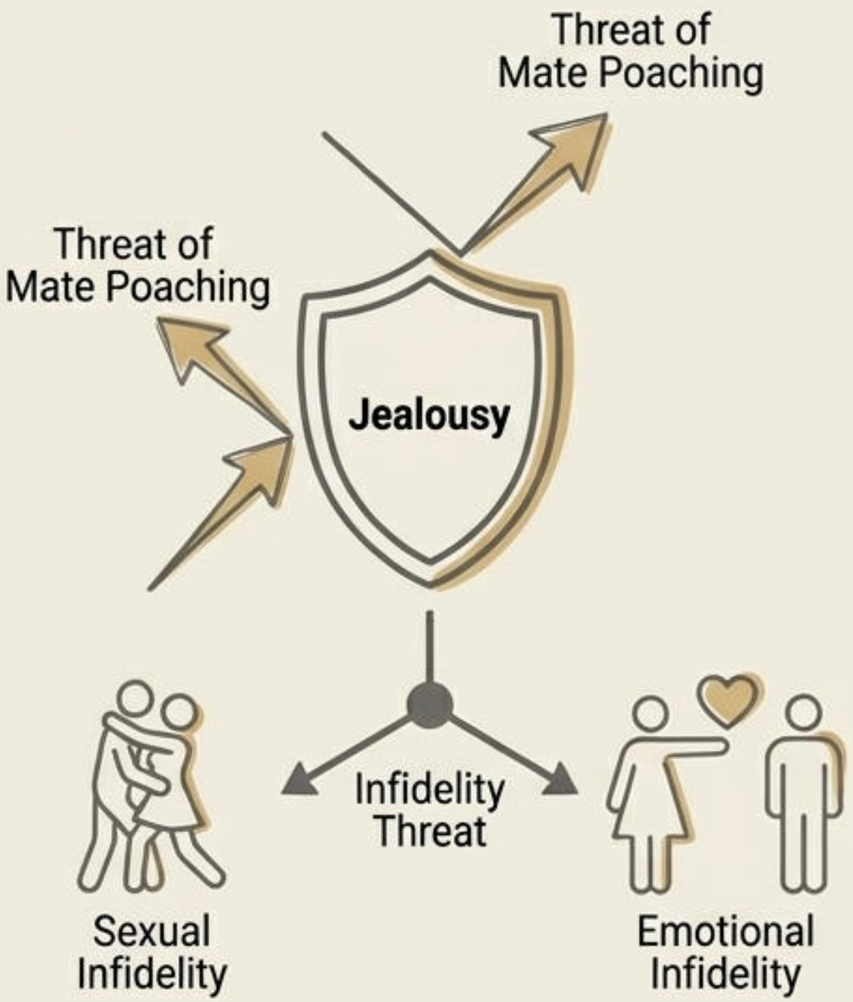
Jealousy: The Evolved ‘Guard’ Against Poaching

Jealousy is the evolved psychological mechanism designed to combat the threat of mate poaching and relationship desertion. It is a negative emotional state triggered by the actual or suspected loss of a partner’s sexual services or affection. Its function is to motivate behaviors that protect the pair-bond from intruders and prevent the loss of a valuable reproductive partner.

Two Distinct Forms of Infidelity Threat:
The threat of infidelity is not monolithic. From an evolutionary perspective, it presents two distinct adaptive problems, which correspond to two types of infidelity:

1. **Sexual Infidelity:** A partner engaging in passionate sexual intercourse with another person. This poses a direct and specific adaptive problem for men.
2. **Emotional Infidelity:** A partner forming a deep emotional attachment and falling in love with another person. This poses a direct and specific adaptive problem for women.



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Sex Differences in Jealousy Triggers

Evolutionary psychology predicts that men and women will find different types of infidelity more distressing because each type threatens a different aspect of their reproductive success.

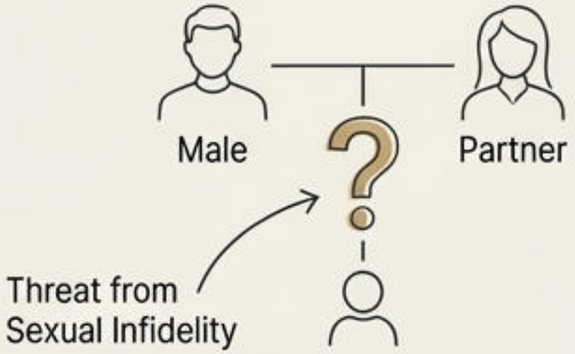
Male Perspective: The Problem of Paternity Uncertainty

- A woman is always 100% certain of her maternity. A man can never be 100% certain of his paternity.
- **Threat:** A female partner's sexual infidelity raises the devastating possibility that he might invest years of resources (time, food, protection) in a child who is not his own, carrying a rival's genes. This is a catastrophic reproductive loss.
- **Prediction:** Men will be more distressed and jealous over a partner's **sexual infidelity**.

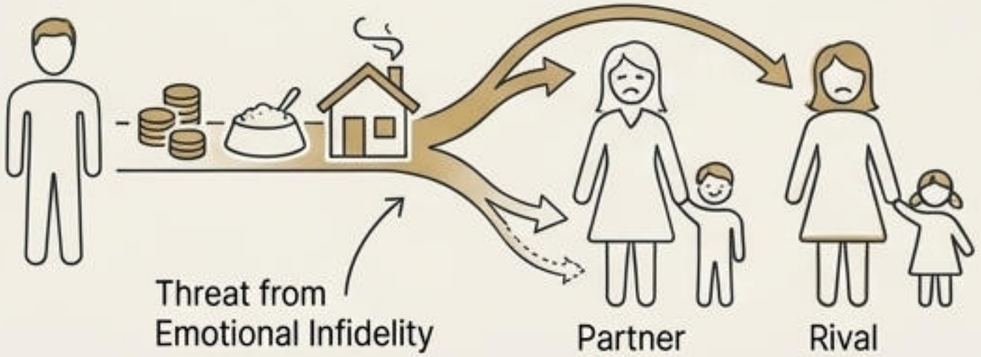
Female Perspective: The Problem of Resource Diversion

- A woman's primary concern is securing the continued investment of her partner to help raise her offspring to maturity.
- **Threat:** A male partner's emotional infidelity—falling in love with another woman—signals a potential long-term diversion of his time, commitment, and resources to a rival and her children. This threatens the survival and success of her own offspring.
- **Prediction:** Women will be more distressed and jealous over a partner's **emotional infidelity**.

Male Concern: Paternity Uncertainty



Female Concern: Resource Diversion



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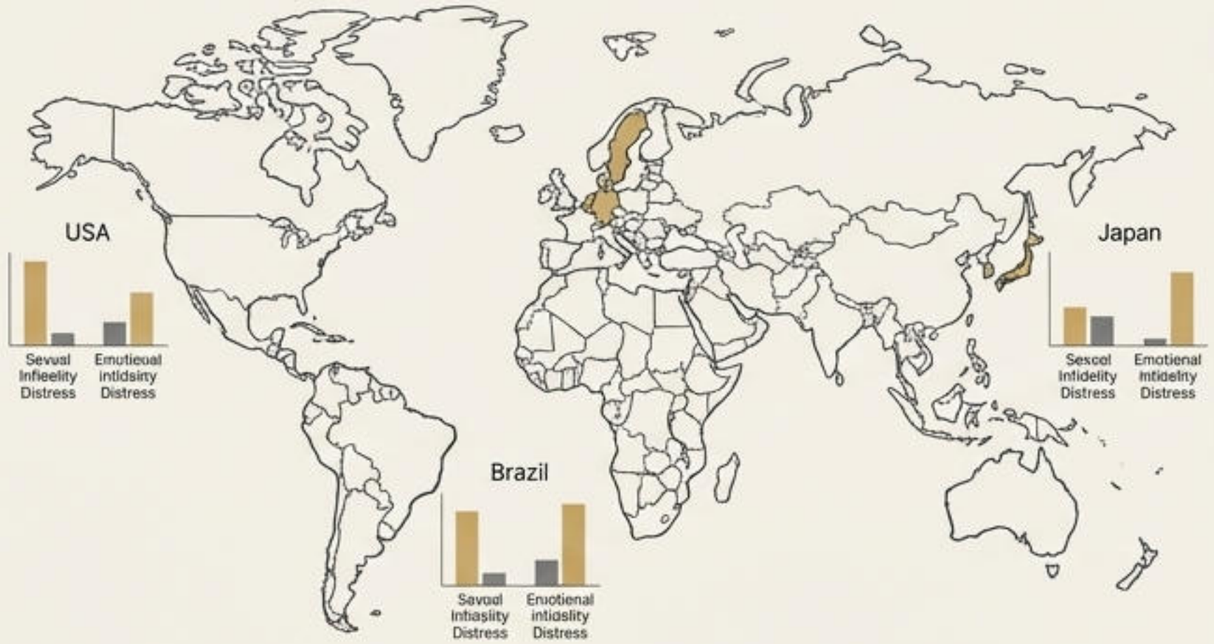
Cross-Cultural Evidence for Sex Differences in Jealousy

The predicted sex difference in jealousy triggers has been one of the most robustly tested and supported hypotheses in evolutionary psychology. Numerous studies have asked individuals to **choose which scenario would be more upsetting: their partner having passionate sex with someone else, or their partner forming a deep emotional bond with someone else.**

Consistent Global Findings:
Across a wide array of countries—including the United States, Germany, the Netherlands, Brazil, Japan, Korea, and Sweden—the results consistently show:

- **Men** report significantly more distress to imagining **sexual infidelity**.
- **Women** report significantly more distress to imagining **emotional infidelity**.

This consistent pattern across diverse cultural contexts provides strong evidence that this sex difference is a fundamental, evolved feature of human psychology, designed to solve the distinct adaptive problems faced by males and females in long-term relationships.



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